

Submission answers

"Explain how well you implemented the task, and why is it the best possible way to do so?"

The task. Extend the MVP with a capability about limits, conditions and non-negotiable requirements — showing whether the related work is complete, partial, blocked or unresolved — that fits the existing product and stays independent of any theme.

What we built. A constraint gate that sits in front of every irreversible action. It returns one of the four required states, with precedence `BLOCKED > UNRESOLVED > PARTIAL > COMPLETE`, and it enforces two invariants everywhere: *unresolved is never a pass*, and *you cannot waive a rule that was never measured*.

Why this is the right design rather than one of several.

1. **The engine knows nothing about the domain; rules are data.** This is the only structure that makes theme independence provable rather than claimed. The same binary evaluates orbital manoeuvres, a software release gate and an FAA aircraft-dispatch rulebook, with zero code changes — and a test asserts the engine source contains no domain vocabulary at all. An engine with domain logic could only ever *assert* portability.
2. **Non-negotiable means an absence of code, not a policy flag.** Rules marked `waivable: false` have no override branch anywhere. This cannot be misconfigured, escalated past, or forgotten under pressure, which a permission model cannot promise.
3. **Failure is loud by construction.** A rule that throws becomes `UNEVALUATED`, never `SATISFIED`. A rule whose applicability is unknown is `UNEVALUATED`, not skipped. Most gates fail open under exactly these conditions.
4. **Every dead end carries its exit.** Each unresolved rule names the measurement that would clear it, and the gate computes the cheapest sufficient set — with minimality verified by re-evaluating every proper subset — or proves the state terminal, in which case it recommends nothing. A four-state signal that cannot say what to do next is a diagnosis without a treatment.

Evidence it works, rather than assurances.

- **Formal:** a TLA+ specification model-checked with TLC over 148,163 states and eight invariants, including a two-state counterexample that reproduces a real bug we shipped and fixed.
- **Property-based:** eight properties over thousands of generated rulebooks — among them *COMPLETE implies every rule was actually evaluated*, and *a rule that throws never becomes SATISFIED*. 224 assertions pass in total.
- **External:** our satellite-vision layer was validated against a population dataset it never saw

(Spearman +0.75), and its failure band above 60°N was measured and encoded as a refusal rather than hidden.

- **Reproducible:** every decision emits an RFC 8785 canonical-JSON, hash-chained receipt that replays byte-for-byte.

It fits the MVP because it constrains what the MVP already did. Round 1 planned manoeuvres and voted on them. Round 2 puts the gate in front of that vote — the API itself returns 409 when unauthorised, so the interface cannot route around it.

What we are careful not to claim. The model check is bounded, not a proof for unbounded inputs. Minimality is verified exhaustively over a small declared evidence universe, not proved over a logical encoding. Propellant balances are operator-declared and flagged as such in every record, and one rule — maritime clearance — is permanently unevaluable because we have no AIS feed, which we left visible on purpose. A system arguing that unknowns must be reported would be a poor advertisement for itself if it hid its own.

"What makes it innovative?"

The novel part is not the four states — those are deliberately isomorphic to OASIS XACML's Permit / Deny / Indeterminate / NotApplicable, because inventing decision semantics is how projects get them wrong. Three things are new:

1. **The same rule returns a different state depending on the quality of evidence available.** Positional knowledge is read from SpaceX's public Starlink ephemerides where they exist — 3 m at closest approach, which passes the standard — and from our own TLE-derived conformal bound where they do not — 1,689 m, which fails it. Nothing about the satellite changed; what changed is how much is known about it.
2. **Two provably different kinds of "unresolved."** One is fixable with data, and the system states how much is needed, from the coverage law for split conformal prediction. The other is impossible in principle, and it cites why. Telling an operator to collect more data when nothing can help wastes a mission.
3. **A measured finding against the incumbent method.** Casualty expectancy is computed on a real 2-D population field and beside the 1-D latitude-band method the regulatory tool uses. Across 18,257 sampled footprints the 1-D method understates exposure 94.1% of the time over the densest ground, by a median factor of 3.9. That is a falsifiable result, not a feature.